

Unsupervised Learning

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Unsupervised Learning

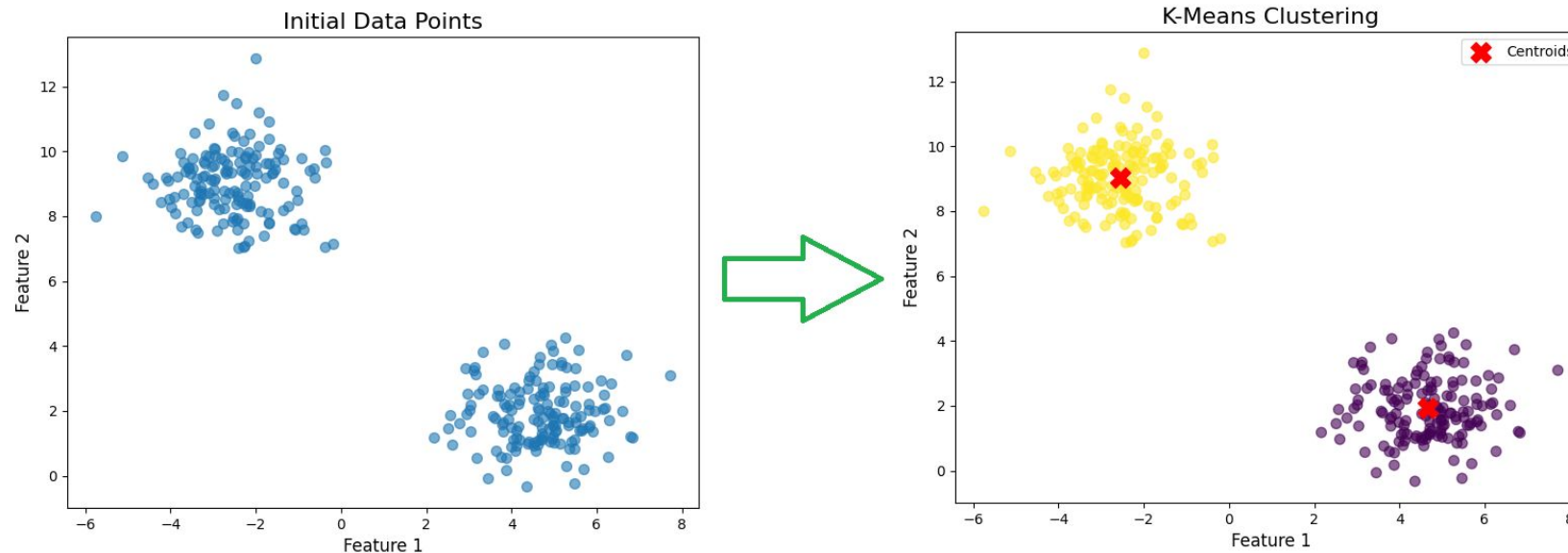
A type of learning where the model works with unlabeled data to discover patterns.

Examples:

- Clustering (e.g., K-Means)
- Dimensionality Reduction (e.g., PCA)

Clustering

- A way of grouping the data points into different clusters, consisting of similar data points
- Objects with possible similarities remain in a group that has less or no similarities with another group



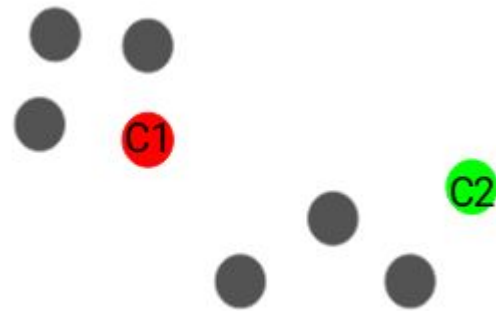
K-means Clustering

- K-means clustering is a method for grouping n observations into K clusters
- Aims to assign each observation to the cluster with the nearest mean or centroid
- We have these 8 points, and we want to apply k-means to create clusters for these points

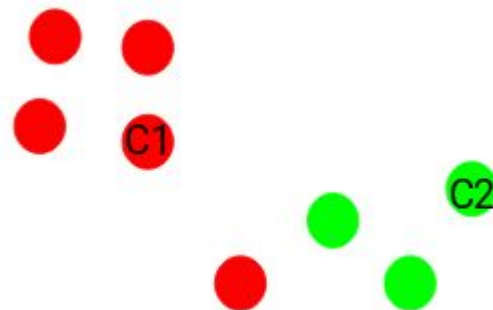


Intuition

1. Choose the number of clusters k . Let's say it is 2.
2. Select k random points from the data as centroids

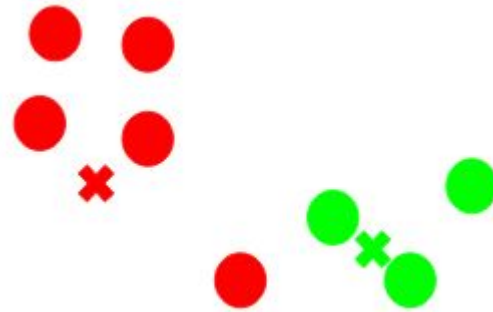


3. Assign all the points to the closest cluster centroid

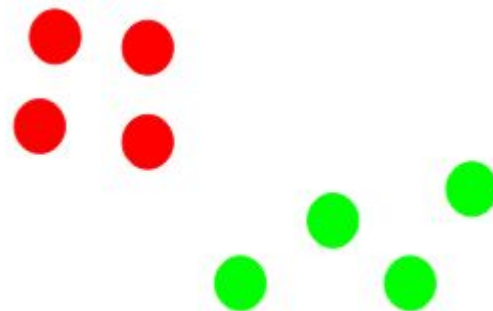


Intuition (Cont.)

4. Recompute the centroids of newly formed clusters



5. Repeat steps 3 and 4



Stopping Criteria

- Centroids of newly formed clusters do not change
- Points remain in the same cluster
- Maximum number of iterations is reached

Challenges

- Value of k must be predefined
 - A wrong value of k might result in inaccurate result
 - Optimal k can be found using silhouette analysis or elbow method
- The algorithm cannot handle categorical data
 - Distance calculation is inaccurate with categorical features
- High computational cost
- Finds spherical shaped clusters only

Example with Data

- Let's consider the following data points.
- Point (x, y) A (1,2) B (2,3) C (3,3) D (6,7) E (7,8) F (8,8)
- **Step 1: Choose the number of clusters (k)**
 - We decide to divide the dataset into 2 clusters, $k = 2$
- **Step 2: Select Initial Centroids Randomly**
 - Centroid 1 (C1) = A (1,2)
 - Centroid 2 (C2) = E (7,8)

Example with Data

- **Step 3: Assign Each Point to the Nearest Centroid**

- Using Euclidean distance
- Cluster C1: { A(1,2), B(2,3), C(3,3) }
- Cluster C2: { D(6,7), E(7,8), F(8,8) }

- **Step 4: Recompute Centroids**

- The new centroid is the mean of all points in the cluster:
- $C1 = (\frac{1+2+3}{3}, \frac{2+3+3}{3}) = (2, 2.67)$
- $C2 = (\frac{6+7+8}{3}, \frac{7+8+8}{3}) = (7, 7.67)$

Example with Data

- **Step 5: Repeat Steps 3 & 4 Until Convergence**
 - Final Clusters:
 - Cluster 1 (C1): A(1,2), B(2,3), C(3,3)
 - Cluster 2 (C2): D(6,7), E(7,8), F(8,8)

Questions?